

RAMCO INSTITUTE OF TECHNOLOGY

Department of Mechanical Engineering

Academic Year: 2020-2021 (Odd Semester)

Innovative Practices- Virtual Laboratory Session Handling Details

Name of the Faculty	: Mr. T.Selva Sundar & Mr.K.Amudhan
Subject Code and Title	: ME8791 Mechatronics
Year and Branch	: IV year Mechanical
Date	: 01.10.2020
Topics Covered	: Programmable Logic Controllers- Timer and Counters
Name of the Laboratory	: Industrial Electric Drives and Application of PLC
Website Link	: http://ied-nitk.vlabs.ac.in/index.html

All PLC's have timer instructions. Timers are output instructions that are internal to the programmable logic controller. Timers provide timed control of the devices that they activate or de-activate.

Basic functions of timer

- Timers are used to delay an action.
- Timers are used to run an operation for a predetermined period of time.
- Timers are also used to record the total accumulated time of continuous or intermediate events.

Basic functions of counter

A counter is a PLC instruction that either increments (counts up) or decrements (counts down) an integer number value when prompted by the transition of a bit from 0 to 1 ("false" to "true").

Counter instructions come in three basic types:

1. up counters,
2. down counters, and
3. Up/down counters.

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Simulator

Legend:
Instruction is true (Green)
Instruction is false (Red)

Input module: Start (I:1/0), Reset (I:1/1)

PLC

Output module: Motor ON (O:2/1), Motor OFF (O:2/2), Done Timing (O:2/3)

Motor (0.24)

Ladder Logic Diagram:

- 000: Start (NO) → RTO (T42)
- 001: Timer Enable (NO) → RTO (T42)
- 002: Timer Timingbit (NO) → RTO (T42)
- 003: Timer Timingbit (NC) → RTO (T42)
- 004: Done Timing (NO) → RTO (T42)
- 005: Reset (NO) → RTO (T42)

RTO (Retentive Timer On) Parameters:

- Timer: T42
- Time Base: 1.0
- Preset: 50
- Accum: 0

Output Status:

- Motor ON (O:2/1)
- Motor OFF (O:2/2)
- Done Timing (O:2/3)

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